

Locate the forced and natural poles, or select circuit parameters to place the natural poles at specified locations.
Examples 10-1, 10-2, 10-3, 10-4, 10-5, 10-6, 10-7, 10-8, and Exercises 10-1, 10-2, 10-3, 10-4, 10-5, 10-6, 10-7, 10-8, 10-9, 10-10, 10-11

10-11 The switch in Figure P10-11 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$ and $i_L(t)$ in symbolic form.

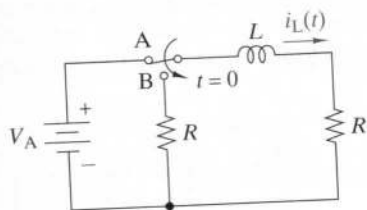


FIGURE P10-11

10-12 The switch in Figure P10-11 has been in position B for a long time and is moved to position A at $t = 0$. Transform the circuit into the s domain and solve for $I_L(s)$ and $i_L(t)$ in symbolic form.

10-13 The switch in Figure P10-13 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_C(s)$ and $i_C(t)$ in symbolic form.

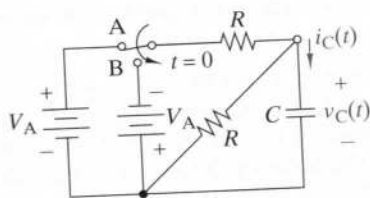


FIGURE P10-13

10-14 The switch in Figure P10-13 has been in position B for a long time and is moved to position A at $t = 0$. Transform the circuit into the s domain and solve for $V_C(s)$ and $v_C(t)$ in symbolic form.

10-15 Transform the circuit in Figure P10-15 into the s domain and find $I_L(s)$, $i_L(t)$, $V_L(s)$, and $v_L(t)$ when $v_1(t) = V_A u(t)$ and $i_L(0) = I_A$.

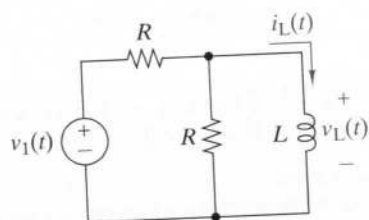


FIGURE P10-15

10-16 Transform the circuit in Figure P10-15 into the s domain and find $I_L(s)$, $i_L(t)$, $V_L(s)$, and $v_L(t)$ when $v_1(t) = V_A e^{-at} u(t)$ and $i_L(0) = I_A$.

10-17 The switch in Figure P10-17 has been in position A for a long time and is moved to position B at $t = 0$.

(a) Transform the circuit into the s domain and solve for $I_L(s)$ in symbolic form.

(b) Repeat part (a) using MATLAB.

(c) Find $i_L(t)$ for $R_1 = R_2 = 500 \Omega$, $R_3 = 1 \text{ k}\Omega$, $L = 250 \text{ mH}$, $C = 4 \mu\text{F}$, and $V_A = 15 \text{ V}$.

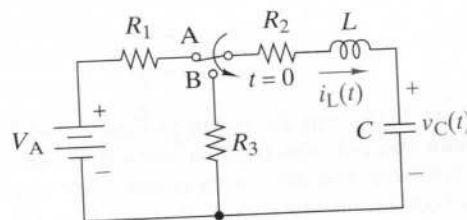


FIGURE P10-17

10-18 The switch in Figure P10-17 has been in position B for a long time and is moved to position A at $t = 0$.

(a) Transform the circuit into the s domain and solve for $V_C(s)$ in symbolic form.

(b) Find $v_C(t)$ for $R_1 = R_2 = 250 \Omega$, $R_3 = 1 \text{ k}\Omega$, $L = 500 \text{ mH}$, $C = 1 \mu\text{F}$, and $V_A = 15 \text{ V}$.

(c) Repeat part (b) using OrCAD.

10-19 The circuit in Figure P10-19 is in the zero state. The s -domain relationship between the input $I_1(s)$ and the output $I_R(s)$ is usually given as a ratio called a *network function*. Find $I_R(s)/I_1(s)$.

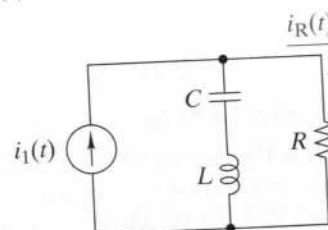


FIGURE P10-19

10-20 The circuit in Figure P10-20 is in the zero state. Find the s -domain relationship between the input $I_1(s)$ and the output $V_O(s)$.

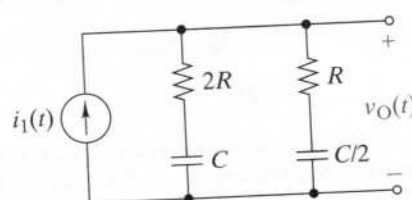


FIGURE P10-20

- 10-21 The initial conditions in Figure P10-21 are $v_C(0) = 0$ and $i_L(0) = I_0$. Transform the circuit into the s domain and use superposition and voltage division to find the zero-state and zero-input components of $V_C(s)$.

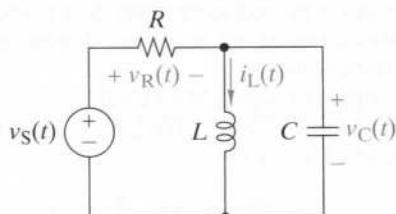


FIGURE P10-21

- 10-22 (a) Transform the circuit in Figure P10-21 into the s domain and use superposition and voltage division to find the zero-state and zero-input components of $V_R(s)$. The initial conditions are $v_C(0) = 0$ and $i_L(0) = I_0$.
 (b) Solve for $v_R(t)$ using OrCAD if $R = 1.5 \text{ k}\Omega$, $L = 500 \text{ mH}$, $C = 1 \text{ }\mu\text{F}$, $v_S(t) = 15u(t) \text{ V}$, $i_L(0) = 10 \text{ mA}$, and $v_C(0) = 0$.

- 10-23 There is no energy stored in the capacitor in Figure P10-23 at $t = 0$. Transform the circuit into the s domain and use current division to find $v_O(t)$ when the input is $i_S(t) = 2.5e^{-200t}u(t) \text{ mA}$. Identify the forced and natural poles in $V_O(s)$.

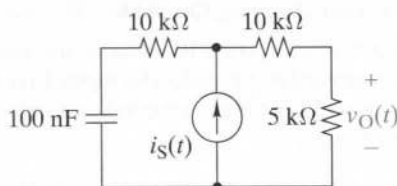


FIGURE P10-23

- 10-24 Repeat Problem 10-23 for $i_S(t) = 2.5e^{-400t}u(t) \text{ mA}$.

- 10-25 The circuit in Figure P10-25 is in the zero state. Use a Thévenin equivalent to find the s -domain relationship between the input $I_S(s)$ and the interface current $I(s)$.

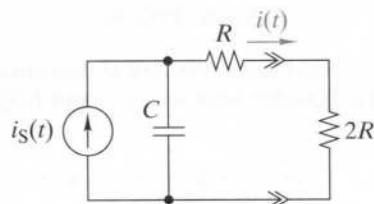


FIGURE P10-25

- 10-26 Find the Thévenin equivalent circuit seen by the $2R$ load resistor in the circuit of Figure 10-25 when $v_C(0) = V_0$.
 10-27 The circuit in Figure P10-27 is in the zero state. Find the Thévenin equivalent to the left of the interface.

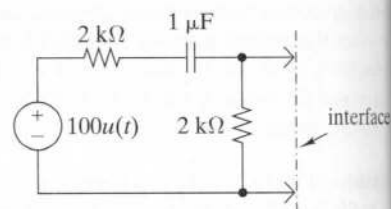


FIGURE P10-27

- 10-28 There is no initial energy stored in the circuit in Figure P10-28. Transform the circuit into the s domain and use superposition to find $V(s)$. Identify the forced and natural poles in $V(s)$.

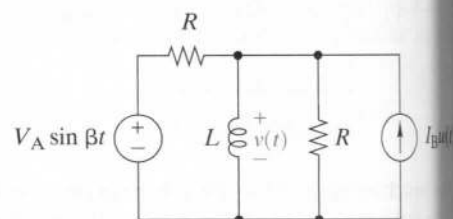


FIGURE P10-28

- 10-29 The equivalent impedance between a pair of terminals is

$$Z_{EQ}(s) = 1000 \frac{s + 2000}{s + 1000}$$

A voltage $v(t) = 10u(t)$ is applied across the terminals. Find the resulting current response $i(t)$.

- 10-30 There is no initial energy stored in the circuit in Figure P10-30. Use circuit reduction to find the network function $V_2(s)/V_1(s)$. Then select values of R and C so that the poles of the network function are approximately -2618 and -382 rad/s .

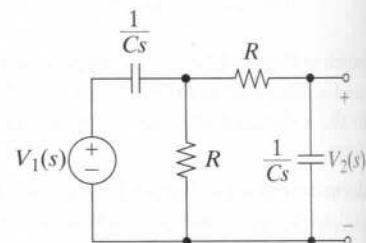


FIGURE P10-30

OBJECTIVE 10-3 GENERAL CIRCUIT ANALYSIS (SECTS. 10-4, 10-5, 10-6)

Given a linear circuit:

- (a) Determine the initial conditions (if not given) and transform the circuit into the s domain.

- Solve for zero-state and/or zero-input response transforms and waveforms using node-voltage or mesh-current methods.
- Identify the forced and natural poles or select circuit parameters to place the natural poles at specified locations.

Examples 10-10, 10-11, 10-13, 10-14, 10-15, 10-16, 10-17 and Exercises 10-12, 10-13, 10-14, 10-15, 10-16, 10-17, 10-18, 10-19, 10-20, 10-21

10-31 There is no initial energy stored in the circuit in Figure P10-31.

- Transform the circuit into the s domain and formulate mesh-current equations.
- Solve these equations for $I_2(s)$ in symbolic form.
- Identify the poles and zeros of $I_2(s)$.
- Find $i_2(t)$ for $v_1(t) = 50u(t)$ V, $R_1 = 1$ k Ω , $R_2 = 2$ k Ω , $L = 2$ H, and $C = 500$ nF.

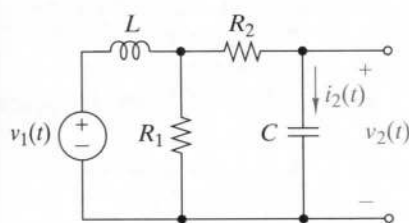


FIGURE P10-31

10-32 There is no initial energy stored in the circuit in Figure P10-32.

- Transform the circuit into the s domain and formulate node-voltage equations.
- Solve these equations for $V_2(s)$ in symbolic form.
- Identify the natural and forced poles of $V_2(s)$.
- Find $v_2(t)$ for $v_1(t) = 15u(t)$ V, $R_1 = R_2 = 500$ Ω , $L = 15$ H, and $C = 2$ μ F.

10-33 There is no initial energy stored in the circuit in Figure P10-33.

- Transform the circuit into the s domain and formulate node-voltage equations.
- Solve these equations for $V_2(s)$ in symbolic form.
- Insert a buffer at point A and solve for $V_2(s)$ in symbolic form. How did inserting the buffer change the denominator and, therefore, the location of the poles?

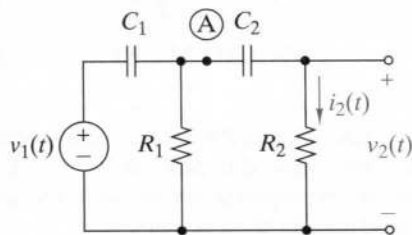


FIGURE P10-33

10-34 There is no initial energy stored in the circuit in Figure P10-33. The Thévenin equivalent circuit to the left of point A when a unit step is applied is

$$V_T(s) = \frac{1}{s + 10^3} \text{ V-s, and } Z_T(s) = \frac{10^6}{s + 10^3} \Omega$$

Select values for R_2 and C_2 such that the output transform is

$$V_2(s) = \frac{s}{s^2 + 3000s + 10^6} \text{ V-s}$$

10-35 There is no initial energy stored in the bridged-T circuit in Figure P10-35.

- Transform the circuit into the s domain and formulate mesh-current equations.
- Use the mesh-current equations to find the s -domain relationship between the input $V_1(s)$ and the output $V_2(s)$.

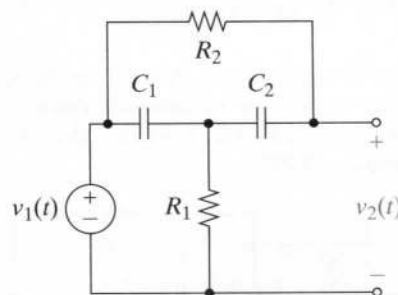


FIGURE P10-35

10-36 There is no initial energy stored in the bridged-T circuit in Figure P10-35.

- Transform the circuit into the s domain and formulate node-voltage equations.
- Use the node-voltage equations to find the s -domain relationship between the input $V_1(s)$ and the output $V_2(s)$.

10-37 There is no initial energy stored in the circuit in Figure P10-37.

- Find the zero-state mesh currents $i_A(t)$ and $i_B(t)$ when $v_1(t) = 100e^{-200t}u(t)$ V.
- Validate your answers using OrCAD.

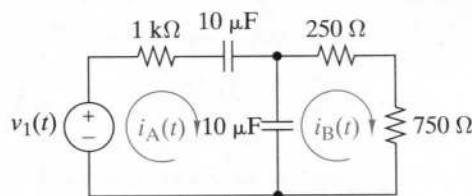


FIGURE P10-37

10-38 There is no external input in the circuit in Figure P10-38.